Gradient Descent

An Analytical and Numerical Example

Gradient

DreamStudio: A bowling ball rolling down a hill

Our first step is to go over a detailed calculation of a gradient, which is at the basis of NN optimizers

Gradient – Detailed Calculation

$$L = \frac{1}{m} * \sum_{n=1}^{m} \left(y_{predicted,n} - y_n\right)^2$$

- What does this function represent?

Gradient – Detailed Calculation

$$L = \frac{1}{m} * \sum_{n=1}^{m} \left(y_{predicted,n} - y_n\right)^2$$

- This is the Mean Squared Error (MSE) Loss Function, common in regression problems

Gradient – Detailed Calculation

$$L = \frac{1}{m} * \sum_{n=1}^{m} \left(y_{predicted,n} - y_n\right)^2$$

- What does each parameter represent?

Gradient – Detailed Calculation

$$L = \frac{1}{m} * \sum_{n=1}^{m} \left(y_{predicted,n} - y_n\right)^2$$

Number of data instances

The prediction by the model. The model’s output

Ground truth value for instance *n*

Gradient – Detailed Calculation

$$L = \frac{1}{m} * \sum_{n=1}^{m} (y_{predicted,n} - y_n)^2$$

$$y_{predicted} = w0 + w_1 * x_1$$

Let’s assume a linear regression model, with two parameters: an intercept (w_0) and slope (w_1). The equation above represents the output of this linear model, given an input (x_1)

Gradient – Detailed Calculation

$$L = \frac{1}{m} * \sum_{n=1}^{m} (y_{predicted,n} - y_n)^2$$

$$y_{predicted} = w0 + w_1 * x_1$$

What will be the gradient of the loss function with respect to w_0 and w_1 ?

Gradient – Detailed Calculation

$$L = \frac{1}{m} * \sum_{n=1}^{m} (y_{predicted,n} - y_n)^2$$

$$y_{predicted} = w0 + w_1 * x_1$$

What will be the gradient of the loss function with respect to w_0 and w_1 ?

$$\left(\frac{\partial(L)}{\partial(w_0)}, \frac{\partial(L)}{\partial(w_1)}\right)$$

It is a vector with two elements – the partial derivatives of the loss function with respect to each of the trainable parameters

Gradient – Detailed Calculation

$$L = \frac{1}{m} * \sum_{n=1}^{m} (y_{predicted,n} - y_n)^2$$

$$y_{predicted} = w0 + w_1 * x_1$$

$$\left(\frac{\partial(L)}{\partial(w_0)}, \frac{\partial(L)}{\partial(w_1)}\right)$$

Let’s calculate the value of the first element:

$$\frac{\partial(L)}{\partial(w_0)} = 2 * (w0 + w_1 * x_1 - y)$$

Gradient – Detailed Calculation

$$L = \frac{1}{m} * \sum_{n=1}^{m} (y_{predicted,n} - y_n)^2$$

$$y_{predicted} = w0 + w_1 * x_1$$

$$\left(\frac{\partial(L)}{\partial(w_0)}, \frac{\partial(L)}{\partial(w_1)}\right)$$

$$\frac{\partial(L)}{\partial(w_0)} = 2 * (w0 + w_1 * x_1 - y)$$

And the second element:

$$\frac{\partial(L)}{\partial(w_1)} = 2 * (w0 + w_1 * x_1 - y) * x_1$$

Gradient – Detailed Calculation

So the gradient of the loss function:

$$L = \frac{1}{m} * \sum_{n=1}^{m} (y_{predicted,n} - y_n)^2$$

$$y_{predicted} = w0 + w_1 * x_1$$

Is:

$$\nabla L(w_0, w_1) = (\frac{\partial(L)}{\partial(w_0)}, \frac{\partial(L)}{\partial(w_1)})$$

$$\frac{\partial(L)}{\partial(w_0)} = 2 * (w0 + w_1 * x_1 - y)$$

$$\frac{\partial(L)}{\partial(w_1)} = 2 * (w0 + w_1 * x_1 - y) * x_1$$

Gradient – Detailed Calculation

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$$\frac{\partial(L)}{\partial(w_0)} = 2 * (w0 + w_1 * x_1 - y)$$

$$\frac{\partial(L)}{\partial(w_1)} = 2 * (w0 + w_1 * x_1 - y) * x_1$$

The **functional form of the gradient** is constant – it doesn't change with a change of the input (x,y) or the value of the weights. However the **value of the gradient** will be different for each combination of inputs and weight values.

Gradient – Detailed Calculation

$$\nabla L(w_0, w_1) = \left(\frac{\partial(L)}{\partial(w_0)}, \frac{\partial(L)}{\partial(w_1)}\right)$$

$$\frac{\partial(L)}{\partial(w_0)} = 2 * (w0 + w_1 * x_1 - y)$$

$$\frac{\partial(L)}{\partial(w_1)} = 2 * (w0 + w_1 * x_1 - y) * x_1$$

What will be the value of the gradient for (x=2, y=1, w_0=3, w_1=4)?

Gradient – Detailed Calculation

$$\nabla L(w_0, w_1) = \left(\frac{\partial(L)}{\partial(w_0)}, \frac{\partial(L)}{\partial(w_1)}\right)$$

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(20, 40)

Gradient – Detailed Calculation

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What will be the value of the gradient for (x=2, y=1, w_0=3, w_1=4)?

(20, 40)

The gradient represents the direction in which the loss function will increase the most, for a small step taken in (w_0, w_1) around the point (x,y,w_0,w_1).
The *negative* of the gradient is the direction in (w_0,w_1) space in which the function will decrease most in value, for a small step around the point (x,y,w_0,w_1)